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COLOR MATCHING FOR A PRINTER

Abstract

Systems and methods for color matching. In an embodiment, a method comprises loading a color sample file of a color sample that includes color measurements, and generating a patch chart image by processing the color measurements for the color sample to select target colors, and generating the patch chart image comprising color patch objects corresponding with the target colors. The method further comprises printing the patch chart image at a target printer to output a patch chart, scanning the patch chart to generate color measurements for the patch chart, generating a color matching profile for the target printer based on the color measurements for the color sample and the color measurements for the patch chart, and registering the color matching profile at the target printer.

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Background/Summary

TECHNICAL FIELD

[0001] The following disclosure relates to the field of printing, and in particular, to color management in printing.

BACKGROUND

[0002] Print systems transform a print job into a printed output by applying colorant(s) to mark a printable medium, such as paper. In order to represent colors specified in the print job, a printer may utilize different colorants that are applied to the medium in different ratios. For instance, a CMYK (Cyan, Magenta, Yellow, Key black) printer utilizes combinations of four different colorants to represent the input colors specified in the print job as output colors. The output colors (also referred to as the color space) producible by printers may vary for a variety of reasons, such as bit depth, temperature, types of colorants used, etc.

[0003] Printers, which are not calibrated from a color perspective, may reproduce different results while printing the same digital document. To level the colors onto printers, color calibration techniques may be applied by creating and using International Color Consortium (ICC) color profiles specific for each printer. Even when calibration of a single printer could be performed targeting different goals, some print shops may want to save as much as possible on colored inks (e.g., maximizing the use of black as much as possible), while others may be interested in color fidelity, etc. Tuning the colors and leveling them across the print shop is difficult and complex. In the scenarios of having a physical sample printed in unknown conditions/printer configuration, such as when printed by a different print shop, the task of replicating the same results onto a different printer may be performed using a color matching technique and a spectrometer device. In a present color matching technique, the physical sample printed in unknown conditions (e.g., different print shop), referred to as a “golden sample”, is scanned and colors of the physical sample are measured with a spectrometer device. The same digital document is printed using a target printer and the output (referred to as a “before-matching sample”) is scanned and colors of the before-matching sample are measured using the same spectrometer device. The colors between the golden sample and the before-matching sample are compared and a delta color difference calculated, resulting in a correctional ICC profile specific for the job. Using the correctional ICC profile, the printer can replicate the colors of the golden sample.

[0004] One potential issue is it may be difficult to map the scanned version of the golden sample to the scanned version of the before-matching sample to perform the comparison. For example, the golden sample and/or the before-matching sample may be shifted or rotated when scanning, there may be differences in size and/or resolution of the color samples, etc. Thus, it is desirable to identify improved color matching processes for print systems.

SUMMARY

[0005] Embodiments described herein provide enhanced color matching. As a general overview, a color sample is obtained, which represents the benchmark for color matching (e.g., a golden sample). The color sample is scanned to generate color measurements for the color sample, and the color measurements for the color sample are processed to identify or select target colors in the color sample to be used in color matching. A patch chart image is generated from the target colors, and is printed at a target printer to output a patch chart. The printed patch chart is then scanned (e.g., with a spectrometer device or the like) to generate color measurements for the patch chart. The color measurements for the patch chart and the color measurements for the color sample are compared to generate a color matching profile for the target printer. The color matching profile may then be registered at the target printer for color-matched printing. One technical benefit is the color matching process focuses on matching to the colors found in the color sample, rather than

performing pixel matching between two scanned images (e.g., a scanned version of the golden sample and a scanned version of the before-matching sample). Therefore, the color matching process is more tolerant to scan errors, size or resolution differences, or other conditions that may exist, and may provide more accurate color matching.

[0006] In an embodiment, a method of color matching is disclosed. The method comprises loading a color sample file of a color sample, where the color sample file includes color measurements for the color sample. The method further comprises generating a patch chart image by processing the color measurements for the color sample to select target colors from the color sample, and generating the patch chart image comprising color patch objects corresponding with the target colors. The method further comprises printing the patch chart image at a target printer to output a patch chart that includes a plurality of color patches corresponding with the color patch objects, scanning the patch chart to generate color measurements for the patch chart, generating a color matching profile for the target printer based on the color measurements for the color sample and the color measurements for the patch chart, and registering the color matching profile at the target printer for color-matched printing at the target printer.

[0007] In an embodiment, a color matching apparatus comprises at least one processor and memory. The at least one processor is configured to cause the color matching apparatus at least to receive a color sample file of a color sample, where the color sample file includes color measurements for the color sample. The at least one processor is configured to cause the color matching apparatus at least to generate a patch chart image by processing the color measurements for the color sample to select target colors from the color sample, and generating the patch chart image comprising color patch objects corresponding with the target colors. The at least one processor is configured to cause the color matching apparatus at least to provide an instruction to print the patch image at a target printer to output a patch chart, where the patch chart includes a plurality of color patches corresponding with the color patch objects. The at least one processor is configured to cause the color matching apparatus at least to provide an instruction to scan the patch chart to generate color measurements for the patch chart, receive the color measurements for the patch chart, generate a color matching profile for the target printer based on the color measurements for the color sample and the color measurements for the patch chart, and register the color matching profile at the target printer for color-matched printing at the target printer.

[0008] Other embodiments may include computer readable media, other systems, or other methods as described below.

[0009] The above summary provides a basic understanding of some aspects of the specification. This summary is not an extensive overview of the specification. It is intended to neither identify key or critical elements of the specification nor delineate any scope particular embodiments of the specification, or any scope of the claims. Its sole purpose is to present some concepts of the specification in a simplified form as a prelude to the more detailed description that is presented later.

Description

DESCRIPTION OF THE DRAWINGS

[0010] Some embodiments of the present disclosure are now described, by way of example only, and with reference to the accompanying drawings. The same reference number represents the same element or the same type of element on all drawings.

[0011] FIG. 1 is a diagram of a print system in an illustrative embodiment.

[0012] FIG. 2 is a schematic diagram of an image forming apparatus in an illustrative embodiment.

[0013] FIG. 3 is a block diagram of a management server in an illustrative embodiment.

[0014] FIG. 4 is a flow chart illustrating a method of performing color matching in an illustrative

embodiment.

[0015] FIG. **5** illustrates a color sample.

[0016] FIG. **6** is a schematic diagram illustrating a color matching unit loading a color sample file in an illustrative embodiment.

[0017] FIG. **7** is a schematic diagram illustrating a color matching unit generating a patch chart image in an illustrative embodiment.

[0018] FIG. **8** is a schematic diagram illustrating printing of a patch chart image in an illustrative embodiment.

[0019] FIG. **9** is a schematic diagram illustrating scanning of a patch chart in an illustrative embodiment.

[0020] FIG. **10** is a schematic diagram illustrating a color matching unit generating a color matching profile in an illustrative embodiment.

[0021] FIG. **11** is a schematic diagram illustrating a color matching unit registering a color matching profile at a target printer in an illustrative embodiment.

[0022] FIGS. **12A-12C** are flow charts illustrating a method of generating a patch chart image in an illustrative embodiment.

[0023] FIG. **13** is a diagram illustrating a sample color list and a target color list in an illustrative embodiment.

[0024] FIG. **14** is a diagram illustrating a patch chart image in an illustrative embodiment.

[0025] FIG. **15** is a diagram illustrating a sample color list and a target color list in an illustrative embodiment.

[0026] FIG. **16** is a diagram illustrating a patch chart in an illustrative embodiment.

[0027] FIG. **17** is a flow chart illustrating a method of color matching within a color matching unit in an illustrative embodiment.

[0028] FIG. **18** is a diagram illustrating a color matching screen in an illustrative embodiment.

[0029] FIG. **19** is a diagram illustrating a color matching screen in an illustrative embodiment.

[0030] FIG. **20** is a diagram illustrating a color matching screen in an illustrative embodiment.

[0031] FIG. **21** is a diagram illustrating a color matching screen in an illustrative embodiment.

[0032] FIG. **22** illustrates a processing system operable to execute a computer readable medium embodying programmed instructions to perform desired functions in an illustrative embodiment.

DETAILED DESCRIPTION

[0033] The figures and the following description illustrate specific exemplary embodiments. It will thus be appreciated that those skilled in the art will be able to devise various arrangements that, although not explicitly described or shown herein, embody the principles of the embodiments and are included within the scope of the embodiments. Furthermore, any examples described herein are intended to aid in understanding the principles of the embodiments, and are to be construed as being without limitation to such specifically recited examples and conditions. As a result, the inventive concept(s) is not limited to the specific embodiments or examples described below, but by the claims and their equivalents.

[0034] FIG. **1** is a diagram of a print system **100** in an illustrative embodiment. As one example in FIG. **1**, print system **100** may include an image forming apparatus **102**, a client terminal **112** (e.g., a personal computer (PC)), a management server **114** (also referred to as a print server), and a scanner **116**. As illustrated in FIG. **1**, one or more of the devices of print system **100** are able to perform data communication with one another via a network **110**. The network **110** may comprise, for example, a network including a local area network (LAN), a wide area network (WAN), such as the Internet, etc., and may comprise a wired network, a wireless network, or a network including both of a wired network and a wireless network.

[0035] Image forming apparatus **102** includes a digital front end (DFE) **104** and a printer **106**. In an example, printer **106** may comprise an apparatus that performs image formation (printing) on a recording medium by applying colorants on the basis of print data received from the DFE **104**.

DFE **104** is an information processing apparatus that receives a print job (e.g., from the client terminal **112** or management server **114**), generates print data by a raster image processor (RIP) engine on the basis of the print job, and transmits the print data to the printer **106**. In an embodiment, DFE **104** may be on-ground or on-premises with printer **106**. In general, “on-premises” means that the infrastructure exists on-site in contrast to being hosted off-site. DFE **104** may be implemented on a separate (on-premises) platform from printer **106**, or may be integrated on a platform of the printer **106**. In an embodiment, DFE **104** may communicate with printer **106** over network **110**.

[0036] The client terminal **112** is an information processing apparatus that generates a print job to be printed by a user, and transmits the print job to the DFE **104** or the management server **114**. The management server **114** is a server apparatus that manages the print job received from the client terminal **112**, and transmits the print job to the DFE **104**, such as in response to a request from the DFE **104**. Scanner **116** is a color measurement apparatus that performs a read operation on a recording medium on which a printed image is formed to capture and evaluate colors on the printed image, and to output color measurements for the colors on the printed image. For example, scanner **116** may comprise a spectrophotometric color measuring device (referred to generally as a spectrometer device) configured to capture and evaluate colors on the printed image. Scanner **116** may be directly connected to management server **114**, may be connected to management server **114** over network **110**, or may be connected to other devices.

[0037] FIG. **2** is a schematic diagram of image forming apparatus **102** in an illustrative embodiment. Image forming apparatus **102** is a type of device that executes an image forming process (e.g., printing) on a recording medium. In an embodiment, image forming apparatus **102** includes DFE **104**, and the printer **106** comprising one or more print engines **220**. DFE **104** comprises an apparatus, device, circuitry, means, and/or other component configured to accept a print job **211**, and convert the print job **211** into a suitable format for print engine **220**. DFE **104** includes an Input/Output (I/O) interface **212**, a print controller **214**, and a print engine interface **216**, and may also include a user interface **218**. I/O interface **212** comprises an apparatus, device, circuitry, means, and/or other component configured to receive a print job **211** from a source, such as a client terminal **112**, a management server **114**, etc. I/O interface **212** may be considered a network interface in some embodiments. The print job **211** comprises one or more job files or vector files formatted with a Page Description Language (PDL), such as PostScript, Printer Command Language (PCL), Intelligent Printer Data Stream (IPDS), etc. The print job **211** may also comprise a job ticket containing instructions, requirements, and/or other control information for processing and/or printing a print job, such as a Job Definition Format (JDF) job ticket. Print controller **214** comprises an apparatus, device, circuitry, means, and/or other component configured to transform the print job **211** into print data **219** comprising one or more digital images that may be used by print engine **220** to mark a recording medium **232** with ink, toner, or another recording or marking material. In an embodiment, print controller **214** includes a Raster Image Processor (RIP) **215** that translates or rasterizes the print job **211** to generate digital images or raster images that a printer can understand and print. A digital or raster image comprises a two-dimensional array of pixels or dots, also referred to as a bitmap. Whereas the job file(s) of the print job **211** in PDL format is a high-level description of the content (e.g., text, graphics, pictures, etc.), a digital image defines a pixel value or color value for each pixel in a display space. Print engine interface **216** comprises an apparatus, device, circuitry, means, and/or other component configured to communicate with print engine **220**, such as to transmit digital images to print engine **220**. Print engine interface **216** may be communicatively coupled to print engine **220** via one or more communication links **217** (e.g., a fiber link, a bus, a communication cable, etc.), and is configured to transfer the digital images to print engine **220**. User interface **218** is a component configured to interact with a human operator. A human operator may access user interface **218** to view status indicators or updates, view or manipulate settings, schedule print jobs, etc.

[0038] Printer **106** may comprise a cut-sheet printer, a continuous-form printer that prints on a web of continuous-form media, a wide format printer, etc. Print engine **220** includes a DFE interface **222**, a print engine controller **224**, and a print mechanism **226**. DFE interface **222** comprises an apparatus, device, circuitry, means, and/or other component configured to interact with DFE **104**, such as to receive print data **219** from DFE **104**. Print engine controller **224** comprises an apparatus, device, circuitry, means, and/or other component configured to process the print data **219** (e.g., the digital or raster images) received from DFE **104**, and provide control signals to print mechanism **226**. Print mechanism **226** is an image formation device (or devices) that mark the recording medium **232** with a recording material **234** or colorant, such as ink. Print mechanism **226** may be configured for variable droplet or dot size to reproduce multiple intensity levels. Recording medium **232** comprises any type of material suitable for printing upon which recording material **234** is applied, such as paper (web or cut-sheet), plastic, card stock, transparent sheets, cloth, etc. In an embodiment, print mechanism **226** may include one or more printheads that are configured to jet or eject droplets of a print fluid, such as ink (e.g., water, solvent, oil, or UV-curable), through a plurality of orifices or nozzles. The orifices or nozzles may be grouped according to ink types (e.g., colors such as Cyan (C), Magenta (M), Yellow (Y), Key black (K) or formulas such as for pre-coat, image and protector coat, etc.), which may be referred to as color planes. In another embodiment, print mechanism **226** may include a drum that selectively collects electrically-charged powdered ink (toner), and transfers the toner to recording medium **232**. Media conveyance device **230** may be configured to move recording medium **232** relative to print mechanism **226**. In other embodiments, portions of print mechanism **226** may be configured to move relative to recording medium **232**. Image forming apparatus **102** may include various other components not specifically illustrated in FIG. 2.

[0039] FIG. 3 is a block diagram of management server **114** in an illustrative embodiment. Management server **114** includes or implements a color management unit **310**, which comprises an apparatus, device, circuitry, means, and/or other component configured to perform color management for an image forming apparatus **102** (e.g., printer **106**). In general, color management is a process or procedure of ensuring consistent and/or accurate colors, such as at a printer. Different printers **106** may have different color capabilities and characteristics. Thus, the printed output of a common image produced from different printers **106** may be inconsistent. In image forming apparatuses, color management may be performed to suppress variations in printed output due to device-specific characteristics and enhance the reproducibility of printed output.

[0040] In an embodiment, color management unit **310** may include a color matching unit **316**, which comprises an apparatus, device, circuitry, means, and/or other component configured to perform color matching. Color matching unit **316** is configured to generate a color matching profile **320** (i.e., an ICC profile) for a printer **106** based on a color sample. The color matching profile **320** may then be used in the printer **106** to reproduce the colors of the color sample on printed output as closely as possible.

[0041] In an embodiment, management server **114** may further include a user interface **340**. User interface **340** is an information processing apparatus configured to interact with an end user. For example, user interface **340** may include a display **342**, screen, touch screen, or the like (e.g., a Liquid Crystal Display (LCD), a Light Emitting Diode (LED) display, etc.). User interface **340** may include a keyboard or keypad, a tracking device (e.g., a trackball or trackpad), a speaker, a microphone, etc. Color matching unit **316** may be configured to interact with user interface **340** regarding a color matching process. Thus, user interface **340** may provide or display a Graphical User Interface (GUI) **344** generated by color matching unit **316** for the color matching process, an example of which is further described below.

[0042] Management server **114** may include one or more processors **330** and a memory **332**. Processor **330** represents the internal circuitry, logic, hardware, etc., that provides the functions of management server **114**. Processor **330** may be configured to execute instructions **334** (i.e.,

computer program code) for software that are loaded into memory **332**. Processor **330** may comprise a microprocessor, a set of one or more processors, or may comprise a multi-processor core depending on the particular implementation. Memory **332** is a computer readable storage medium for data, instructions **334**, applications, etc., and is accessible by processor **330**. Memory **332** is a hardware storage device capable of storing information on a temporary basis and/or a permanent basis. Memory **332** may comprise volatile or non-volatile Random-Access Memory (RAM), Read-Only Memory (ROM), FLASH devices, volatile or non-volatile Static RAM (SRAM) devices, magnetic disk drives, Solid State Disks (SSDs), or any other volatile or non-volatile storage device. Management server **114** may include other components or devices not shown in FIG. **3**.

[0043] FIG. **4** is a flow chart illustrating a method **400** of performing color matching in an illustrative embodiment. Method **400** will be discussed with respect to print system **100** of FIG. **1**, although method **400** may be performed by other systems, not shown. The steps of the flow charts described herein may include other steps that are not shown. Also, the steps of the flow charts described herein may be performed in an alternate order.

[0044] In general, method **400** describes an operation workflow for color matching. Color matching operates using a color sample, which is a physical sample (e.g., a preprinted document) representing a desired, preferred, or approved printed output. For example, a color sample may comprise a golden sample, which is used in production processes to represent an approved final sample or ideal representation of a printed product. A color sample that is used for color matching (i.e., represents the benchmark for color matching) may be referred to herein as an “original” color sample. FIG. **5** illustrates a color sample **502**. Color sample **502** is a physical document (i.e., one or more pages or sheets) preprinted with text, images, etc., and comprising a plurality of colors **504** (which may be referred to as “original” colors).

[0045] In FIG. **4**, color matching unit **316** loads a color sample file of the color sample **502** (step **402**). FIG. **6** is a schematic diagram illustrating color matching unit **316** loading a color sample file **604** in an illustrative embodiment. In one example, color sample **502** may have been previously scanned and processed (i.e., color measurement) to generate color sample file **604**, which is stored in memory. Therefore, color sample file **604** may be loaded by selecting the color sample file **604** from a memory location (optional step **414** in FIG. **4**). In another example, color sample **502** may be scanned and processed (e.g., at scanner **116**) to generate color sample file **604** (optional step **416** in FIG. **4**). In this example, the scanner **116** captures and evaluates colors on the color sample **502**, and outputs color measurements contained in the color sample file **604**. Color sample file **604** may include a digital image of the color sample **502** (i.e., color sample image **610**), such as produced by scanning the color sample **502**. Color sample file **604** further includes color measurements **612** generated or computed from the color sample **502**. For example, the color measurements **612** may include CMYK values **614** in the CMYK color space, and/or L*a*b* values **616** in the CIELAB color space, although other types of measurements are considered herein.

[0046] Color matching unit **316** then processes or otherwise evaluates the color measurements **612** of the color sample **502**. In FIG. **4**, color matching unit **316** generates a patch chart image based on the color measurements **612** of the color sample **502** (step **404**). FIG. **7** is a schematic diagram illustrating color matching unit **316** generating a patch chart image **706** in an illustrative embodiment. In general, color sample **502** includes a plurality of colors **504** identifiable from the color measurements **612**. Color matching unit **316** generates a patch chart image **706** comprising color patch objects corresponding with the colors **504** (some or all) identified in the color sample **502**. For example, if Color1 is identified in color sample **502** (e.g., with CMYK value **614**=C1, M1, Y1, K1), then patch chart image **706** may include a color patch object corresponding with Color1. If Color2 is identified in color sample **502** (e.g., with CMYK value **614**=C2, M2, Y2, K2), then patch chart image **706** may include a color patch object corresponding with Color2. A patch chart generation process for generating the patch chart image is described in further detail below. The

patch chart image **706** generated by color matching unit **316** may be referred to as an “original” patch chart image.

[0047] In FIG. **4**, the patch chart image **706** is printed at a target printer **106** to output a patch chart (step **406**). The target printer **106** is the printer upon which color matching is being performed. FIG. **8** is a schematic diagram illustrating printing of a patch chart image **706** in an illustrative embodiment. The patch chart image **706** is sent to the target printer **106**, which in turn prints the patch chart image **706** to generate, produce, or output a patch chart **808**. As will be described in more detail below, patch chart **808** is a physical document (i.e., one or more pages or sheets) that includes a plurality of color patches **810** (printed versions of the color patch objects in the patch chart image **706**). Each color patch **810** represents an actual color **812** (also referred to as a printed color, a reproduced color, etc.) reproduceable by the target printer **106**. For example, if the patch chart image **706** includes a color patch object corresponding with Color1, then the patch chart **808** includes a color patch **810** corresponding with or mapped to Color1 but represents an actual color **812** reproducible by the target printer **106**. If the patch chart image **706** includes a color patch object corresponding with Color2, then the patch chart **808** includes a color patch **810** corresponding with or mapped to Color2 but represents an actual color **812** reproducible by the target printer **106**. Thus, the actual colors **812** printed in the color patches **810** may or may not match corresponding colors **504** of the color sample **502**.

[0048] In FIG. **4**, the printed patch chart **808** is scanned (e.g., at scanner **116**) to generate or compute color measurements for the patch chart **808** (step **408**). FIG. **9** is a schematic diagram illustrating scanning of a patch chart **808** in an illustrative embodiment. For example, the patch chart **808** is loaded on scanner **116**, which scans and processes the patch chart **808** to generate a patch chart file **902** (also referred to as a scanned patch chart file). Scanner **116** captures and evaluates the actual colors **812** on the patch chart **808**, and outputs color measurements for the actual colors **812** on the patch chart **808** (i.e., contained in the patch chart file **902**). Patch chart file **902** may include a digital image of the patch chart **808** (i.e., scanned patch chart image **910**), such as produced by scanning the patch chart **808**. Patch chart file **902** further includes color measurements **912** generated or computed from the patch chart **808**. For example, the color measurements **912** may include CMYK values **914** in the CMYK color space, and/or $L^*a^*b^*$ values **916** in the CIELAB color space, although other types of measurements are considered herein. Patch chart file **902** is then loaded at color matching unit **316**.

[0049] In FIG. **4**, color matching unit **316** generates a color matching profile **320** for the target printer **106** based on the color measurements **612** for the color sample **502** and the color measurements **912** for the patch chart **808** printed at the target printer **106** (step **410**). FIG. **10** is a schematic diagram illustrating color matching unit **316** generating a color matching profile **320** in an illustrative embodiment. For example, color matching unit **316** may compare the $L^*a^*b^*$ values **616** for the color sample **502** and the $L^*a^*b^*$ values **916** for the patch chart **808** to determine deviations between the colors **504** of the color sample **502** and the actual colors **812** of the patch chart **808** printed by the target printer **106**. Color matching unit **316** may then generate or adjust CMYK color curves **1002** for the target printer **106**.

[0050] In FIG. **4**, color matching unit **316** may then register the color matching profile **320** at the target printer **106** for color-matched printing (step **412**). FIG. **11** is a schematic diagram illustrating color matching unit **316** registering the color matching profile **320** at the target printer **106** in an illustrative embodiment. For example, color matching unit **316** may send the color matching profile **320** to the target printer **106**, which is registered or loaded at the target printer **106**, such as at the print engine controller **224** (see FIG. **2**). The target printer **106** may then print using colors that are adjusted based on the color matching profile **320**, which may generally be referred to as color-matched colors **1102**. One assumption is that the target printer **106** allows for an adjustable ICC profile. Print engine controller **224**, for example, may use the color matching profile **320** when rendering digital images for printing at the print mechanism **226**. The color-matched colors **1102**

output by the print mechanism **226** should therefore match the colors **504** of the color sample **502** as closely as possible. Color matching unit **316** may send the color matching profile **320** along with a print job **211** to the target printer **106** as a bundle, or may send the color matching profile **320** and the print job **211** separately.

[0051] One technical benefit is the color-matched colors **1102** output by the target printer **106** (based on the color matching profile **320**) approximate the colors **504** of the color sample **502** as closely as possible by using the color matching process. The color matching process focuses on matching to the colors **504** found in the color sample **502**, rather than performing pixel matching between two scanned images (e.g., a scanned version of the golden sample and a scanned version of the before-matching sample). Therefore, the color matching process is more tolerant to scan errors, size or resolution differences, or other conditions that may exist, and may provide more accurate color matching.

[0052] FIGS. **12A-12C** are flow charts illustrating a method **1200** of generating a patch chart image **706** in an illustrative embodiment. Method **1200** will be discussed with respect to color matching unit **316** of FIG. **3**, although method **1200** may be performed by other systems, not shown.

[0053] In general, method **1200** describes an operation workflow for a patch chart generation process where color matching unit **316** processes or evaluates the color measurements **612** for the color sample **502**. In FIG. **12A**, color matching unit **316** processes the color measurements **612** of the color sample **502** to select target colors from the color sample **502** (step **1202**). A target color comprises a color **504** (or group of colors **504**) in the color sample **502** selected for color matching. Color matching unit **316** may select each of the colors **504** from the color sample **502** as the target colors (optional step **1206**), or may select a subset of the colors **504** from the color sample **502** as target colors (optional step **1208**). FIG. **13** is a diagram illustrating a sample color list **1302** and a target color list **1312** in an illustrative embodiment. Sample color list **1302** represents each color **504** of the color sample **502** identified based on the color measurements **612**, and target color list **1312** represents target colors **1314** selected from the color sample **502** for color matching.

[0054] In FIG. **12A**, color matching unit **316** generates the patch chart image **706** comprising color patch objects corresponding with the target colors **1314** (step **1204**). FIG. **14** is a diagram illustrating a patch chart image **706** in an illustrative embodiment. Patch chart image **706** is illustrated as an array of pixels **1402**, and comprises a plurality of color patch objects **1410** distributed among the pixels **1402**. The color patch objects **1410** correspond with the target colors **1314** selected by color matching unit **316** (i.e., some or all of the colors **504** of the color sample **502**). A color patch object **1410** therefore comprises a contiguous set of pixels or dots assigned the same target color value. In an embodiment, patch chart image **706** comprises one color patch object **1410** per target color **1314**. The color patch objects **1410** may be arranged in rows and columns as indicated in FIG. **14**, however the layout of the color patch objects **1410** may vary as desired. The color patch objects **1410** may each have a size (e.g., greater than 7 mm×7 mm) and geometry (e.g., square or rectangular) that is readable by a scanner **116** or the like. Patch chart image **706** may further include row/column marker objects **1404** to delineate individual rows and columns of color patch objects **1410**.

[0055] Color matching unit **316** may select the target colors **1314** in a variety of ways or according to different criteria or policy, one of which is shown in FIG. **12B**. In FIG. **12B**, color matching unit **316** generates the target color list **1312** based on colors **504** of the color sample **502** identified based on the color measurements **612** (step **1214**). For example, there may be a single appearance or multiple appearances of a color **504** (i.e., a distinct CMYK value **614**) in the color measurements **612**. Thus, color matching unit **316** may initially include (e.g., all) the colors **504** from the color sample **502** as target colors **1314** in the target color list **1312** (optional step **1222** in FIG. **12B**). In FIG. **13**, for example, Color1-Color16 illustrate distinct colors found in color sample **502** that are included as target colors **1314**. There may be appearances of colors **504** that are very close in value in the color measurements **612**. Thus, color matching unit **316** may group multiple colors **504** from

the color sample **502** that are within a threshold color range as a target color **1314** (optional step **1224** in FIG. **12B**). For example, one color **504** may have a CMYK value **614**=C1, M1, Y1, K1, another color **504** may have a CMYK value **614**=C1, M1+1, Y2, K2, and yet another color **504** may have a CMYK value **614**=C1, M1+2, Y2, K2). When these colors **504** are within a threshold color range (e.g., C+/-5, M+/-5, Y+/-5, K+/-5), color matching unit **316** may group these colors **504** together in the target color list **1312** as a single target color **1314**. In FIG. **13**, for example, Color12-Color16 illustrate a group **1322** of colors **504** from the color sample **502** that are grouped or merged into a target color **1314**.

[0056] In FIG. **12B**, color matching unit **316** may filter the target color list **1312** based on filter criteria (step **1216**). In an embodiment, the filter criteria may exclude certain target colors **1314** from the target color list **1312** that are less influential to the color matching process than other colors. For example, color matching unit **316** may exclude pure CMYK white and/or CMYK black from the target colors **1314** in the target color list **1312** (optional step **1226**). In another example, color matching unit **316** may exclude colors **504** that appear less than a threshold value in the color sample **502** (optional step **1228**). For example, color matching unit **316** may count the number of times of each color **504** appears in the color sample **502** based on the color measurements **612**, and may compare the counts to a configurable threshold amount (also referred to as a count threshold, a percentage threshold, etc.). Color matching unit **316** may include colors **504** from the color sample **502** having a count greater than or equal to the threshold amount as target colors **1314**, and may exclude colors **504** from the color sample **502** having a count lower than the threshold amount. For instance, color matching unit **316** may exclude colors **504** having a count less than ten, twenty, fifty, etc., a count representing less than a threshold percentage of overall colors **504**, such as 0.5%, 1%, 2%, etc., or some other metric. Color matching unit **316** may use other filter criteria to filter the target color list **1312**.

[0057] Color matching unit **316** assigns color match values to the target colors **1314** in the target color list **1312** (step **1218**). A color match value for a target color **1314** is the value to be matched when reproducing that target color **1314** at a printer. In FIG. **12C**, for example, for each target color **1314**, color matching unit **316** may perform the following. When there is a single appearance of a target color **1314** in the color sample **502**, the color match value of the target color **1314** is the L*a*b* value **616** associated with the target color **1314** (step **1230**). FIG. **15** is a diagram illustrating the sample color list **1302** and the target color list **1312** in an illustrative embodiment. In this example, Color1-Color3 represent target colors **1314** that appear a single time in the color sample **502**, so the color match value **1518** is the L*a*b* value **616** associated with the target color **1314**.

[0058] In FIG. **12C**, when there are multiple appearances of a target color **1314** in the color sample **502**, color matching unit **316** calculates an average of the L*a*b* values **616** over the multiple appearances of the target color **1314** in the color sample **502** as the color match value **1518** of the target color **1314** (step **1232**). For example, a target color **1314** (having a certain CMYK value **614**) may appear multiple times in the color sample **502**, and the L*a*b* values **616** corresponding with each instance of a target color **1314** may be slightly different. Thus, color matching unit **316** may average the L*a*b* values **616** over the individual instances of the target color **1314** appearing in the color sample **502** to determine an average value. In FIG. **15**, for example, Color4-Color5 and Color7-Color10 represent target colors **1314** that appear multiple times in the color sample **502**, so the color match value **1518** is an average of the L*a*b* values **616** associated with the target color **1314**.

[0059] In FIG. **12C**, when grouping multiple colors **504** from the color sample **502** together into a target color **1314**, color matching unit **316** calculates an average value of the L*a*b* values **616** over the multiple colors **504** as the color match value **1518** for the corresponding target color **1314** (step **1234**). In FIG. **15**, for example, Color12-Color16 are grouped into a single target color **1314** (i.e., Color14). Color matching unit **316** may average the L*a*b* values **616** over the individual

colors **504** in the group **1322** to determine an average value.

[0060] It is also noted that FIG. **15** illustrates some target colors **1314** (i.e., Color6 and Color11) have been removed due to filtering.

[0061] Color matching unit **316** generates the patch chart image **706** comprising color patch objects **1410** corresponding with the target colors **1314** (see step **1204** of FIG. **12A**). The patch chart image **706** is then printed at a target printer **106** to output a patch chart (see step **406** of FIG. **4**). FIG. **16** is a diagram illustrating a patch chart **808** in an illustrative embodiment. Patch chart **808** comprises (physical) printed output from the target printer **106**, such as one or more sheets of paper. Patch chart **808** comprises a plurality of color patches **810** representing actual colors **812** reproduceable by the target printer **106** based on the target colors **1314** identified for the color matching process. In an embodiment, patch chart **808** comprises one color patch **810** per target color **1314**. The color patches **810** may be arranged in rows and columns as indicated in FIG. **16**, however the layout of the color patches **810** may vary as desired. The color patches **810** may each have a size (e.g., greater than 7 mm×7 mm) and geometry (e.g., square or rectangular) that is readable by a scanner **116** or the like. Patch chart **808** may further include row/column markers **1604** to delineate individual rows and columns of color patches **810**.

[0062] FIG. **17** is a flow chart illustrating a method **1700** of color matching within color matching unit **316** in an illustrative embodiment. In general, method **1700** describes an operation workflow for color matching within color matching unit **316**. Color matching unit **316** may provide a website, a portal, a dashboard, GUI **344**, or some other digital or graphical interface configured to interact with a user. For example, a GUI **344** provided by color matching unit **316** may include icons, graphic representations or graphical elements, windows, screens, etc. Color matching unit **316** displays a color matching screen (step **1702**) allowing a user, requestor, or operator to initiate a color matching process. FIG. **18** is a diagram illustrating a color matching screen **1800** in an illustrative embodiment. Color matching screen **1800** displays one or more graphical elements **1802** that allow a user to initiate the color matching process for a target printer **106**.

[0063] In FIG. **17**, color matching unit **316** provides an instruction, command, or request to a user to load a color sample file **604** (step **1704**). FIG. **19** is a diagram illustrating a color matching screen **1900** in an illustrative embodiment. Color matching screen **1900** displays a graphical element **1902** that instructs the user to load a color sample file **604**. In response to the instruction, color matching unit **316** receives the color sample file **604** (step **1706**). As described above, a color sample **502** may have been previously scanned and processed to generate color sample file **604**, which is stored in memory. Therefore, color matching screen **1900** may allow the user to select the color sample file **604** from a memory location (optional step **1722**). In another example, the color sample **502** may be scanned and processed (e.g., at scanner **116**) to generate a color sample file **604**, which is received (e.g., automatically) by color matching unit **316** (optional step **1724**). Color matching screen **1900** may display a thumbnail image of the color sample file **604** selected or display other information regarding the color sample file **604**.

[0064] In FIG. **17**, color matching unit **316** generates a patch chart image **706** based on the color measurements **612** of the color sample **502** (step **1708**). Operations for generating the patch chart image **706** were described above in step **404** of FIG. **4** and will not be repeated for the sake of brevity. Color matching unit **316** provides an instruction, command, or request to the user to print the patch chart image **706** at the target printer **106** (step **1710**). FIG. **20** is a diagram illustrating a color matching screen **2000** in an illustrative embodiment. Color matching screen **2000** displays a graphical element **2002** that instructs the user to print the patch chart image **706**. When the user makes a selection to print the patch chart image **706**, color matching unit **316** may send the patch chart image **706** to the target printer **106**, which in turn prints the patch chart image **706** to generate or output a patch chart **808**.

[0065] In FIG. **17**, color matching unit **316** provides an instruction, command, or request to the user to scan the printed patch chart **808** (step **1712**). FIG. **21** is a diagram illustrating a color

matching screen **2100** in an illustrative embodiment. Color matching screen **2100** displays a graphical element **2102** that instructs the user to scan the printed patch chart **808**. In response to scanning of the printed patch chart **808**, such as at scanner **116**, color matching unit **316** receives a scanned patch chart file **902** (see step **1714** of FIG. **17**). In FIG. **17**, color matching unit **316** receives an instruction, command, or selection from the user to execute color matching (step **1716**). In FIG. **21**, for example, a user may select an execute button **2104** or icon on color matching screen **2100** to execute color matching. Color matching unit **316** then generates a color matching profile **320** for the target printer **106** based on the color measurements **612** for the color sample **502** and the color measurements **912** for the patch chart **808** printed at the target printer **106** (step **1718**). Color matching unit **316** may then register the color matching profile **320** at the target printer **106** for color-matched printing (step **1720**). One technical benefit is color matching unit **316** provides a user-friendly interface to implement color matching for a printer **106**.

[0066] Embodiments disclosed herein can take the form of software, hardware, firmware, or various combinations thereof. FIG. **22** illustrates a processing system **2200** operable to execute a computer readable medium embodying programmed instructions to perform desired functions in an illustrative embodiment. Processing system **2200** is operable to perform the above operations by executing programmed instructions tangibly embodied on computer readable storage medium **2212**. In this regard, embodiments of the invention can take the form of a computer program accessible via computer-readable medium **2212** providing program code for use by a computer or any other instruction execution system. For the purposes of this description, computer readable storage medium **2212** can be anything that can contain or store the program for use by the computer.

[0067] Computer readable storage medium **2212** can be an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor device. Examples of computer readable storage medium **2212** include a solid-state memory, a magnetic tape, a removable computer diskette, a random-access memory (RAM), a read-only memory (ROM), a rigid magnetic disk, and an optical disk. Current examples of optical disks include compact disk-read only memory (CD-ROM), compact disk-read/write (CD-R/W), and DVD.

[0068] Processing system **2200**, being suitable for storing and/or executing the program code, includes at least one processor **2202** coupled to program and data memory **2204** through a system bus **2250**. Program and data memory **2204** can include local memory employed during actual execution of the program code, bulk storage, and cache memories that provide temporary storage of at least some program code and/or data in order to reduce the number of times the code and/or data are retrieved from bulk storage during execution.

[0069] Input/output or I/O devices **2206** (including but not limited to keyboards, displays, pointing devices, etc.) can be coupled either directly or through intervening I/O controllers. Network adapter interfaces **2208** may also be integrated with the system to enable processing system **2200** to become coupled to other data processing systems or storage devices through intervening private or public networks. Modems, cable modems, IBM Channel attachments, SCSI, Fibre Channel, and Ethernet cards are just a few of the currently available types of network or host interface adapters. Display device interface **2210** may be integrated with the system to interface to one or more display devices, such as printing systems and screens for presentation of data generated by processor **2202**.

[0070] The following clauses and/or examples pertain to further embodiments or examples. Specifics in the examples may be used anywhere in one or more embodiments. The various features of the different embodiments or examples may be variously combined with some features included and others excluded to suit a variety of different applications. Examples may include subject matter such as a method, means for performing acts of the method, at least one machine-readable medium including instructions that, when performed by a machine cause the machine to perform acts of the method, or of an apparatus or system according to embodiments and examples described herein.

[0071] Some embodiments pertain to Example 1 that includes a method of color matching. The method comprises loading a color sample file of a color sample, wherein the color sample file includes color measurements for the color sample. The method comprises generating a patch chart image by: processing the color measurements for the color sample to select target colors from the color sample, and generating the patch chart image comprising color patch objects corresponding with the target colors. The method comprises printing the patch chart image at a target printer to output a patch chart that includes a plurality of color patches corresponding with the color patch objects, scanning the patch chart to generate color measurements for the patch chart, generating a color matching profile for the target printer based on the color measurements for the color sample and the color measurements for the patch chart, and registering the color matching profile at the target printer for color-matched printing at the target printer.

[0072] Example 2 includes the subject matter of Example 1, where the patch chart image comprises one color patch object per target color.

[0073] Example 3 includes the subject matter of Examples 1 and 2, where the processing the color measurements for the color sample comprises selecting each color from the color sample as the target colors.

[0074] Example 4 includes the subject matter of Examples 1-3, where the processing the color measurements for the color sample comprises selecting a subset of colors from the color sample as the target colors.

[0075] Example 5 includes the subject matter of Examples 1-4, where the selecting comprises grouping multiple colors of the color sample within a threshold color range as one of the target colors.

[0076] Example 6 includes the subject matter of Example 1-5, further comprising assigning a color match value to the one of the target colors by calculating an average of $L^*a^*b^*$ values over the multiple colors of the color sample.

[0077] Example 7 includes the subject matter of Examples 1-6, where the selecting comprises excluding, from the target colors, colors of the color sample that appear less than a threshold amount in the color sample.

[0078] Example 8 includes the subject matter of Examples 1-7, further comprising assigning a color match value to one of the target colors having multiple appearances in the color sample by calculating an average of $L^*a^*b^*$ values over the multiple appearances.

[0079] Some embodiments pertain to Example 9 that includes a color matching apparatus comprising at least one processor and memory. The at least one processor is configured to cause the color matching apparatus at least to receive a color sample file of a color sample, wherein the color sample file includes color measurements for the color sample. The at least one processor is configured to cause the color matching apparatus at least to generate a patch chart image by processing the color measurements for the color sample to select target colors from the color sample, and generating the patch chart image comprising color patch objects corresponding with the target colors. The at least one processor is configured to cause the color matching apparatus at least to provide an instruction to print the patch image at a target printer to output a patch chart, wherein the patch chart includes a plurality of color patches corresponding with the color patch objects. The at least one processor is configured to cause the color matching apparatus at least to provide an instruction to scan the patch chart to generate color measurements for the patch chart, receive the color measurements for the patch chart, generate a color matching profile for the target printer based on the color measurements for the color sample and the color measurements for the patch chart, and register the color matching profile at the target printer for color-matched printing at the target printer.

[0080] Example 10 includes the subject matter of Example 9, where the patch chart image comprises one color patch object per target color.

[0081] Example 11 includes the subject matter of Examples 9 and 10, where the at least one

processor is further configured to cause the color matching apparatus at least to select a subset of colors from the color sample as the target colors.

[0082] Example 12 includes the subject matter of Examples 9-11, where the at least one processor is further configured to cause the color matching apparatus at least to group multiple colors of the color sample within a threshold color range as one of the target colors.

[0083] Example 13 includes the subject matter of Examples 9-12 where the at least one processor is further configured to cause the color matching apparatus at least to assign a color match value to the one of the target colors by calculating an average of $L^*a^*b^*$ values over the multiple colors of the color sample.

[0084] Example 14 includes the subject matter of Examples 9-13, where the at least one processor is further configured to cause the color matching apparatus at least to exclude, from the target colors, colors of the color sample that appear less than a threshold amount in the color sample.

[0085] Example 15 includes the subject matter of Examples 9-14, where the at least one processor is further configured to cause the color matching apparatus at least to assign a color match value to one of the target colors having multiple appearances in the color sample by calculating an average of $L^*a^*b^*$ values over the multiple appearances.

[0086] Example 16 includes the subject matter of Examples 9-15, where a printing system comprises the color matching apparatus and the target printer.

[0087] Example 17 includes the subject matter of Examples 9-16, where a printing system comprises the color matching apparatus, the target printer, and a scanner configured to scan the patch chart.

[0088] Some embodiments pertain to Example 18 that includes a method of color matching. The method comprises receiving, at a color matching apparatus, a color sample file of a color sample, wherein the color sample file includes color measurements for the color sample. The method comprises generating a patch chart image at the color matching apparatus by processing the color measurements for the color sample to select target colors from the color sample, and generating the patch chart image comprising color patch objects corresponding with the target colors. The method comprises providing, at the color matching apparatus, an instruction to print the patch image at a target printer to output a patch chart, wherein the patch chart includes a plurality of color patches corresponding with the color patch objects. The method comprises providing, at the color matching apparatus, an instruction to scan the patch chart to generate color measurements for the patch chart, receiving, at the color matching apparatus, the color measurements for the patch chart, generating, at the color matching apparatus, a color matching profile for the target printer based on the color measurements for the color sample and the color measurements for the patch chart, and registering the color matching profile at the target printer for color-matched printing at the target printer.

[0089] Example 19 includes the subject matter of Example 18, where the patch chart image comprises one color patch object per target color.

[0090] Example 20 includes the subject matter of Examples 18 and 19, where the processing the color measurements for the color sample comprises selecting a subset of colors from the color sample as the target colors.

[0091] Although specific embodiments were described herein, the scope of the invention is not limited to those specific embodiments. The scope of the invention is defined by the following claims and any equivalents thereof.

Claims

1. A method of color matching, the method comprising: loading a color sample file of a color sample, wherein the color sample file includes color measurements for the color sample; generating a patch chart image by: processing the color measurements for the color sample to select target colors from the color sample; and generating the patch chart image comprising color patch objects

corresponding with the target colors; printing the patch chart image at a target printer to output a patch chart that includes a plurality of color patches corresponding with the color patch objects; scanning the patch chart to generate color measurements for the patch chart; generating a color matching profile for the target printer based on the color measurements for the color sample and the color measurements for the patch chart; and registering the color matching profile at the target printer for color-matched printing at the target printer.

2. The method of claim 1, wherein: the patch chart image comprises one color patch object per target color.

3. The method of claim 1, wherein the processing the color measurements for the color sample comprises: selecting each color from the color sample as the target colors.

4. The method of claim 1, wherein the processing the color measurements for the color sample comprises: selecting a subset of colors from the color sample as the target colors.

5. The method of claim 4, wherein the selecting comprises: grouping multiple colors of the color sample within a threshold color range as one of the target colors.

6. The method of claim 5, further comprising: assigning a color match value to the one of the target colors by calculating an average of $L^*a^*b^*$ values over the multiple colors of the color sample.

7. The method of claim 4, wherein the selecting comprises: excluding, from the target colors, colors of the color sample that appear less than a threshold amount in the color sample.

8. The method of claim 1, further comprising: assigning a color match value to one of the target colors having multiple appearances in the color sample by calculating an average of $L^*a^*b^*$ values over the multiple appearances.

9. A color matching apparatus, comprising: at least one processor and memory, wherein the at least one processor is configured to cause the color matching apparatus at least to: receive a color sample file of a color sample, wherein the color sample file includes color measurements for the color sample; generate a patch chart image by: processing the color measurements for the color sample to select target colors from the color sample; and generating the patch chart image comprising color patch objects corresponding with the target colors; provide an instruction to print the patch image at a target printer to output a patch chart, wherein the patch chart includes a plurality of color patches corresponding with the color patch objects; provide an instruction to scan the patch chart to generate color measurements for the patch chart; receive the color measurements for the patch chart; generate a color matching profile for the target printer based on the color measurements for the color sample and the color measurements for the patch chart; and register the color matching profile at the target printer for color-matched printing at the target printer.

10. The color matching apparatus of claim 9, wherein: the patch chart image comprises one color patch object per target color.

11. The color matching apparatus of claim 9, wherein the at least one processor is further configured to cause the color matching apparatus at least to: select a subset of colors from the color sample as the target colors.

12. The color matching apparatus of claim 11, wherein the at least one processor is further configured to cause the color matching apparatus at least to: group multiple colors of the color sample within a threshold color range as one of the target colors.

13. The color matching apparatus of claim 12, wherein the at least one processor is further configured to cause the color matching apparatus at least to: assign a color match value to the one of the target colors by calculating an average of $L^*a^*b^*$ values over the multiple colors of the color sample.

14. The color matching apparatus of claim 11, wherein the at least one processor is further configured to cause the color matching apparatus at least to: exclude, from the target colors, colors of the color sample that appear less than a threshold amount in the color sample.

15. The color matching apparatus of claim 9, wherein the at least one processor is further configured to cause the color matching apparatus at least to: assign a color match value to one of

the target colors having multiple appearances in the color sample by calculating an average of L*a*b* values over the multiple appearances.

16. A printing system, comprising: the color matching apparatus of claim 9; and the target printer.

17. A printing system, comprising: the color matching apparatus of claim 9; the target printer; and a scanner configured to scan the patch chart.

18. A method of color matching, the method comprising: receiving, at a color matching apparatus, a color sample file of a color sample, wherein the color sample file includes color measurements for the color sample; generating a patch chart image at the color matching apparatus by: processing the color measurements for the color sample to select target colors from the color sample; and generating the patch chart image comprising color patch objects corresponding with the target colors; providing, at the color matching apparatus, an instruction to print the patch image at a target printer to output a patch chart, wherein the patch chart includes a plurality of color patches corresponding with the color patch objects; providing, at the color matching apparatus, an instruction to scan the patch chart to generate color measurements for the patch chart; receiving, at the color matching apparatus, the color measurements for the patch chart; generating, at the color matching apparatus, a color matching profile for the target printer based on the color measurements for the color sample and the color measurements for the patch chart; and registering the color matching profile at the target printer for color-matched printing at the target printer.

19. The method of claim 18, wherein: the patch chart image comprises one color patch object per target color.

20. The method of claim 18, wherein the processing the color measurements for the color sample comprises: selecting a subset of colors from the color sample as the target colors.
